

Homework Set 1

S&DS 5360 | Multivariate Statistics

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Problem 1

We have the vectors:

$$\mathbf{a} = [5 \quad 1 \quad 14 \quad 3 \quad -5 \quad 12 \quad -20]$$

and

$$\mathbf{b} = [0 \quad 2 \quad -4 \quad 10 \quad 4 \quad -3 \quad -4]$$

Which we store in R as:

```
a <- c(5, 1, 14, 3, -5, 12, -20)
b <- c(0, 2, -4, 10, 4, -3, -4)
```

(a) $\mathbf{a} + \mathbf{b}$

```
a+b
```

```
[1] 5 3 10 13 -1 9 -24
```

(b) $3\mathbf{a} - 4\mathbf{b}$

```
3*a - 4*b
```

```
[1] 15 -5 58 -31 -31 48 -44
```

(c) Length of $\mathbf{a}$

In this case, length denotes the Euclidean Norm, or magnitude $\|\mathbf{a}\| = \sqrt{\sum a_i^2}$.

```
sqrt(sum(a^2))
```

```
[1] 28.28427
```

(d) Scalar product $\mathbf{a} \cdot \mathbf{b}$

```
a %*% b
```

```
      [,1]  
[1,]      0
```

(e) Interpretation of (d)

Since the scalar (dot) products of $\mathbf{a}$ and $\mathbf{b}$ is 0, we can conclude that the vectors are orthogonal.

Problem 2

Consider the vector:

$$\mathbf{a} = [10 \quad 6 \quad -5]$$

Which we store in R as:

```
a <- c(10, 6, -5)
```

(a)

Define the three-dimensional standard basis vectors to be $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, and $\mathbf{e}_3 = (0, 0, 1)$. It follows, then, that

$$\mathbf{a} = (10 \times \mathbf{e}_1) + (6 \times \mathbf{e}_2) - (5 \times \mathbf{e}_3)$$

(b)

Recall the geometric definition of the dot product:

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta) \\ \cos(\theta) &= \frac{\mathbf{a} \cdot \mathbf{e}_2}{\|\mathbf{a}\| \|\mathbf{e}_2\|} \\ &= \frac{6}{\|\mathbf{a}\|}\end{aligned}$$

We use R to calculate this to be:

```
6/sqrt(sum(a^2))
```

```
[1] 0.4728662
```

(c)

The subspace spanned by $\mathbf{e}_2$ and $\mathbf{e}_3$ is the yz -plane. The projection of $\mathbf{a}$ onto this plane is therefore $\mathbf{a}_{\text{proj}} = (0, 6, -5)$.

Problem 3

(a)

The resulting matrix is a 1×3 matrix.

(b)

The resulting matrix is a 12×9 matrix.

(c)

These two matrices cannot be multiplied, as the vector $\mathbf{a}$ only has one column, which does not match the three rows of $\mathbf{B}$.

(d)

The resulting matrix is a 1×12 matrix.

Problem 4

Recall that a matrix is invertible if its inverse exists. That is, $\exists \mathbf{A}^{-1}$ s.t. $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, which guarantees that the rows and columns of $\mathbf{A}$ are all linearly independent. For the provided matrix, note that column 3 is the sum of columns 1 and 2. Thus, the matrix is **not invertible**.

Problem 5

We wish to calculate the characteristic equation and eigenvalues for the following matrix:

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Given a constant λ and identity matrix $\mathbf{I}$, define

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 - \lambda & 3 \\ 2 & 4 - \lambda \end{bmatrix}$$

The determinant of this matrix is:

$$|\mathbf{A} - \lambda\mathbf{I}| = (1 - \lambda)(4 - \lambda) - 6$$

Setting this equation to zero and solving for λ yields the characteristic equation:

$$\lambda^2 - 5\lambda - 2 = 0$$

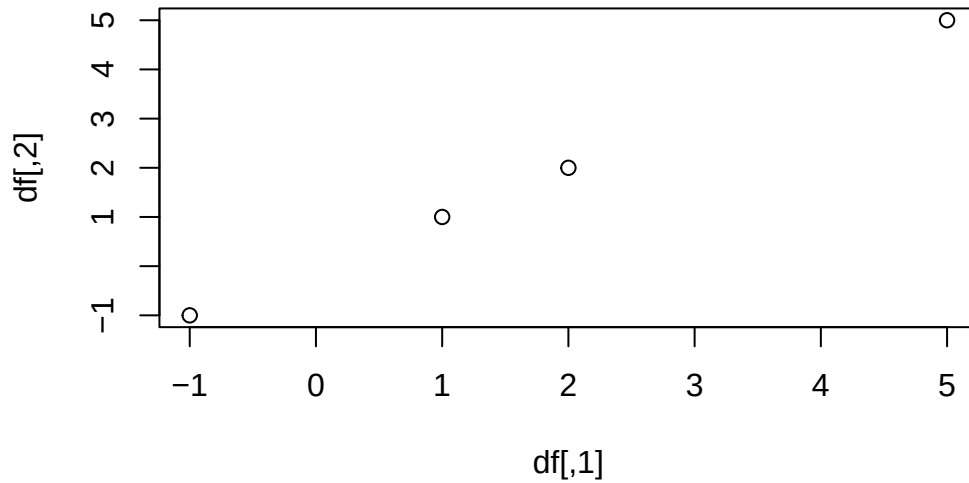
Solving this characteristic equation yields our two eigenvalues:

$$\lambda = \frac{5}{2} \pm \frac{\sqrt{33}}{2} \approx -0.3723 \text{ or } 5.372$$

Problem 6

Consider points $(1, 1), (2, 2), (5, 5), (-1, -1)$. We want to find a rotation of the space that requires only one dimension to describe the points. Note that these points appear to reside along the line $y = x$ with a y -intercept of 0:

```
df <- matrix(c(1, 1, 2, 2, 5, 5, -1, -1), ncol = 2, byrow = TRUE)
plot(df)
```



By this observation, if we **rotate the points by 45 degrees**, the points will all lie in one dimension. Particularly, if we rotate counter-clockwise by -45° , the points will lie along the x -axis.

Define our rotation matrix to be as follows for $\theta = -45^\circ$:

$$R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} \cos(-45^\circ) & -\sin(-45^\circ) \\ \sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix}$$

We rotate the points by multiplying our dataframe by the transpose of the rotation matrix defined above:

```
theta <- -45 * (pi / 180) # Convert to radians.

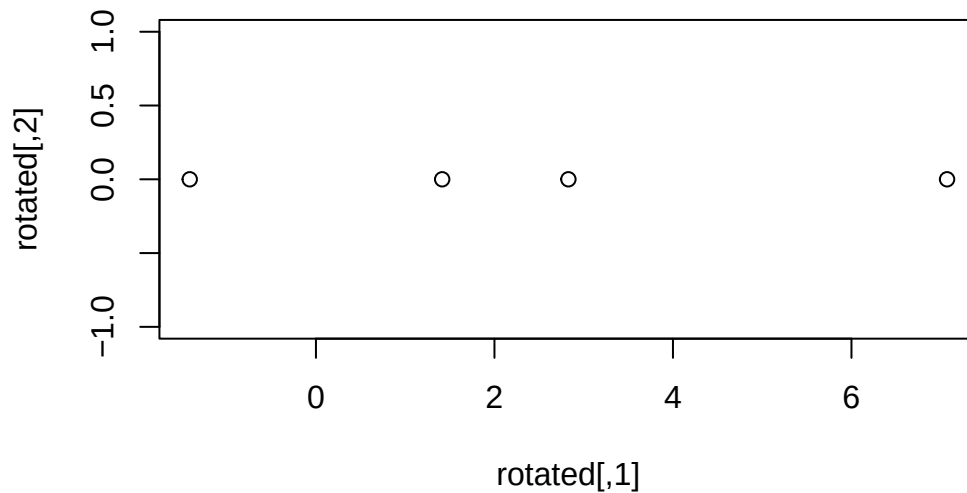
rot_mat <- matrix(c(cos(theta), -sin(theta),
                    sin(theta), cos(theta)),
                  nrow = 2, byrow = TRUE)

rotated <- round(df %*% t(rot_mat), 4); rotated
```

```
      [,1] [,2]
[1,]  1.4142  0
[2,]  2.8284  0
[3,]  7.0711  0
[4,] -1.4142  0
```

Our resulting rotated point matrix is in one dimension, as constructed:

```
plot(rotated)
```



Problem 7

We are provided two matrices, which we encode into R:

```
A <- matrix(c(3,9,12,4,6,0,2,-3,17,-5,3,1,19,19,19,19), nrow = 4, byrow = TRUE)
B <- matrix(c(132,130,156,210,255,255,255,0,130,132,156,1,130,130,1,9),
            nrow = 4, byrow = TRUE)
```

(a) Transpose of B .

```
t(B)
```

	[,1]	[,2]	[,3]	[,4]
[1,]	132	255	130	130
[2,]	130	255	132	130
[3,]	156	255	156	1
[4,]	210	0	1	9

(b) Product of AB .

```
A %% B
```

```
      [,1] [,2] [,3] [,4]
[1,]  4771  4789  4639  678
[2,]   662   654  1245 1235
[3,]  1489  1461  1846 3582
[4,] 12293 12293 10792 4180
```

(c) Inverse of A .

```
solve(A)
```

```
      [,1]      [,2]      [,3]      [,4]
[1,] -0.066420664  0.1070111 -0.001845018  0.03097689
[2,] -0.114391144  0.2398524 -0.142066421  0.06943096
[3,]  0.188191882 -0.1365314  0.088560886 -0.06583803
[4,] -0.007380074 -0.2103321  0.055350554  0.01806176
```

(d) Determinant of A .

```
det(A)
```

```
[1] -10298
```

(e) AA^{-1} .

```
A %% solve(A)
```

```
      [,1]      [,2]      [,3]      [,4]
[1,] 1.000000e+00  0.000000e+00  1.387779e-16 -5.551115e-17
[2,] -4.857226e-17  1.000000e+00  0.000000e+00  4.163336e-17
[3,] -3.348016e-16 -5.551115e-16  1.000000e+00  1.804112e-16
[4,]  8.326673e-17 -4.440892e-16  0.000000e+00  1.000000e+00
```

(f) Trace of B .

```
sum(diag(B))
```

```
[1] 552
```

(g) Trace of B^T .

```
sum(diag(t(B)))
```

```
[1] 552
```